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SARDAR PATEL UNIVERSITY

Bachelor of Computer Application (Semester 5) Examination – 2025

Subject Code: US05MABCA01 || Computer Networks

Date: 04/10/2025 (Saturday)

Time: 10:00 AM to 12:30 PM

Total: 50 Marks

Q-1 Multiple Choice Questions

[10]

1. What does networking software consume on each computer?
a) Power supply b) Memory space
c) CPU speed d) Internet data
2. Which type of network spans across countries or continents?
a) LAN b) MAN
c) PAN d) WAN
3. A ____ is a set of devices referred to as nodes connected by media links.
a) Network b) Internet
c) Intranet d) Extranet
4. Which medium can eliminate most crosstalk effectively?
a) UTP b) Optical fiber
c) STP d) Coaxial
5. What is the shape of twisted-pair wires?
a) Straight b) Circular
c) Helical d) Flat
6. An artificial satellite needs to have _____ which is the path in which it travels around the Earth.
a) Channel b) Tower
c) Footprint d) Orbit
7. A ____ is a set of rules that govern data communications.
a) Layers b) Network
c) Protocol d) Communication
8. The signal from a satellite is normally aimed at a specific area called _____.
a) Channel b) Tower
c) Footprint d) Orbit
9. A switch performs filtering on the basis of _____ address.
a) MAC b) Logical
c) Port d) Postal
10. A Bluetooth Network is called a _____.
a) Piconet b) Scatternet
c) Wireless d) None of these

Q-2 List and explain advantages and disadvantages of computer networks. [10]

OR

Q-2 What is Computer Network? Explain three categories of Computer Network in detail. [10]

Q-3 Describe the construction, types, and uses of twisted pair cables. Explain the differences between Shielded Twisted Pair (STP) and Unshielded Twisted Pair (UTP) cables. [10]

OR

Q-3 What is Multiplexing? Explain Frequency Division Multiplexing (FDM), Time Division Multiplexing (TDM) with appropriate diagrams. [10]

Q-4 Write a note on satellite Network. [10]

OR

Q-4 What is OSI Model? Explain any three layers of OSI Model in detail along with diagram. [10]

Q-5 Describe Modem, Repeater, Switch, Router and Bridge in detail. [10]

OR

Q-5 Write short note on CSMA/CD. [10]
